

Corrigendum to ‘Accelerating regularized tensor decomposition using the alternating direction method of multipliers with multiple Nesterov’s extrapolations’ [Signal Processing 222 (2024) 109532]

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The authors regret that the regularization item $\mathcal{R}(\mathbf{U}^{(n)})$ in equation (4) is lost in the original article during text typing. The correct form of equation (4) is

$$\begin{aligned} \mathcal{L}_\rho \left(\mathbf{A}^{(n)}, \mathbf{U}^{(n)}, \mathbf{\Lambda}^{(n)} \right) = & \frac{1}{2} \left\| \mathbf{X}_{(n)} - \mathbf{A}^{(n)} (\mathbf{B}^{(n)})^T \right\|_F^2 + \mathcal{R}(\mathbf{U}^{(n)}) \\ & + \left\langle \mathbf{\Lambda}^{(n)}, \mathbf{A}^{(n)} - \mathbf{U}^{(n)} \right\rangle + \frac{\rho_n}{2} \left\| \mathbf{A}^{(n)} - \mathbf{U}^{(n)} \right\|_F^2. \end{aligned} \quad (4)$$

The authors regret that the denominator ρ_n under the parameter β_n is lost in equations (29) and (30) in the original article during text typing.

The correct form of equation (29) is

$$\begin{aligned} \mathbf{U}^{(n)} = & \begin{cases} \mathbf{A}^{(n)} + \frac{1}{\rho_n} \mathbf{\Lambda}^{(n)} + \frac{\beta_n}{\rho_n} \mathbf{E}, & \text{if } \mathbf{A}^{(n)} + \frac{1}{\rho_n} \mathbf{\Lambda}^{(n)} < -\frac{\beta_n}{\rho_n} \mathbf{E}; \\ 0, & \text{if } \left| \mathbf{A}^{(n)} + \frac{1}{\rho_n} \mathbf{\Lambda}^{(n)} \right| \leq \frac{\beta_n}{\rho_n} \mathbf{E}; \\ \mathbf{A}^{(n)} + \frac{1}{\rho_n} \mathbf{\Lambda}^{(n)} - \frac{\beta_n}{\rho_n} \mathbf{E}, & \text{if } \mathbf{A}^{(n)} + \frac{1}{\rho_n} \mathbf{\Lambda}^{(n)} > \frac{\beta_n}{\rho_n} \mathbf{E}; \end{cases} \\ & = \text{sign} \left(\mathbf{A}^{(n)} + \frac{1}{\rho_n} \mathbf{\Lambda}^{(n)} \right) \cdot \\ & \quad \max \left\{ 0, \left| \mathbf{A}^{(n)} + \frac{1}{\rho_n} \mathbf{\Lambda}^{(n)} \right| - \frac{\beta_n}{\rho_n} \mathbf{E} \right\}, \end{aligned} \quad (29)$$

where $\mathbf{E} \in \mathbb{R}^{I_n \times R}$ is a matrix of all ones.

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The correct form of equation (30) is

$$\begin{aligned} \boldsymbol{U}_k^{(n)} = & \text{sign}\left(\boldsymbol{A}_k^{(n)} + \frac{1}{\rho_n} \widehat{\boldsymbol{\Lambda}}_{k-1}^{(n)}\right) \cdot \\ & \max\left\{0, \left|\boldsymbol{A}_k^{(n)} + \frac{1}{\rho_n} \widehat{\boldsymbol{\Lambda}}_{k-1}^{(n)}\right| - \frac{\beta_n}{\rho_n} \boldsymbol{E}\right\}. \end{aligned} \tag{30}$$

The authors would like to apologise for any inconvenience caused.